

Answer the Following Questions

1. What is number system? List its different types.

Answer:

A number system is a method of writing and representing numbers using digits or symbols.

Different types of number systems are:

1. Decimal number system
2. Binary number system
3. Octal number system
4. Hexadecimal number system

2. Write short notes on:

(i) Decimal Number System

The decimal number system uses ten digits from 0 to 9. Its base is 10. It is the most commonly used number system in daily life.

(ii) Binary Number System

The binary number system uses only two digits: 0 and 1. Its base is 2. Computers understand and process data in binary form.

(iii) Octal Number System

The octal number system uses eight digits from 0 to 7. Its base is 8. It is used as a shorter form of binary numbers.

(iv) Hexadecimal Number System

The hexadecimal number system uses sixteen symbols: 0–9 and A–F. Its base is 16. It is widely used in computers and programming.

3. What is the importance of binary number system in computers?

Answer:

The binary number system is important because computers can understand only two states: ON and OFF. These states are represented by 1 and 0. Therefore, all data and instructions in a computer are stored and processed in binary form.

4. Write the steps to convert a decimal number into binary number. Explain with an example.

Answer:

Steps:

1. Divide the decimal number by 2.
2. Write the remainder.
3. Continue dividing the quotient by 2 until the quotient becomes 0.
4. Write the remainders from bottom to top.

Example: Convert into binary.

$$35_{10} = (?)_2$$

$$35 \div 2 = 17 \text{ remainder } 1$$

$$17 \div 2 = 8 \text{ remainder } 1$$

$$8 \div 2 = 4 \text{ remainder } 0$$

$$4 \div 2 = 2 \text{ remainder } 0$$

$$2 \div 2 = 1 \text{ remainder } 0$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Therefore,

$$35_{10} = 100011_2$$

5. Write the steps to convert a binary number into decimal number. Explain with an example.

Answer:

Steps:

- 1) Write the place values of binary digits as powers of 2.
- 2) Multiply each digit by its place value.
- 3) Add all the results.

Example: Convert binary into decimal.

$$1011_2 = (?)_{10}$$

$$(1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$= 8 + 0 + 2 + 1$$

$$= 11_{10}$$